Reg. No.															

B.Tech. DEGREE EXAMINATION, MAY 2023

Second Semester

21MES101T – ENGINEERING MECHANICS

(For the candidates admitted from the academic year 2022-2023)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part – B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

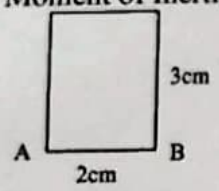
Answer **ALL** Questions

	Marks	BL	CO	PO
1. Mechanics is a branch of science which deals with (A) Failure Analysis (B) Condition of rest or motion of bodies (C) Study of bodies under rest only (D) Study of bodies under motion only	1	1	1	1
2. Action of pull or push of a body at rest or motion is (A) Force (B) Moment (C) Momentum (D) Work	1	1	1	1
3. Two forces of magnitude 5 kN and 8 kN acts vertically upwards and separated by 4m. The resultant will be (A) 3 kN at a distance of 2.46m from 8KN force (B) 13 kN at the middle of two forces (C) 13 kN at a distance of 2.46m from 8 kN force (D) 13 kN at a distance of 2.46m from 5 kN force	1	1	1	2
4. Varignon's theorem of moments states that if a number of coplanar forces acting on a particle are in equilibrium, then (A) Their algebraic sum is zero (B) Their line of action are at equal distances (C) The algebraic sum of their moments about any point is equal to the moment of their resultant force about the same point (D) The algebraic sum of their moments about any point in their plane is zero.	1	1	1	1
5. Frictional force is independent of (A) Type of surface (B) Area of contact (C) Normal Force (D) Coefficient of friction	1	1	2	1
6. When there is no relative motion between contact surfaces a _____ force is developed? (A) Dry friction (B) Dynamic Friction (C) Fluid Friction (D) Static Friction	1	1	2	1
7. In a Screw Jack, the effort required to lower the load is _____ the effort required to raise the same load (A) Equal to (B) More than (C) Less than (D) Un predictable	1	1	2	1
8. The trusses are loaded at _____ (A) Mid point of the truss member (B) Both joints and midpoint of the truss member (C) Joins only (D) Supports only	1	2	2	1
9. Centroid of a quadrant of a circle is given by (A) $\overline{X}=\frac{4r}{3\pi}\ \overline{Y}=\frac{2r}{3\pi}$ (B) $\overline{X}=\frac{4r}{3\pi}\ \overline{Y}=\frac{4r}{3\pi}$ (C) $\overline{X}=\frac{2r}{3\pi}\ \overline{Y}=\frac{4r}{3\pi}$ (D) $\overline{X}=\frac{2r}{3\pi}\ \overline{Y}=\frac{2r}{3\pi}$	1	2	3	1

10. Radius of gyration of a circular plate of diameter 10cm is 1 2 3 2
(A) 1.5 cm (B) 1.75 cm
(C) 2.5 cm (D) 2.25 cm

11. Moment of inertia of a sphere of radius R and mass M about an axis tangential to it is 1 2 3 1
(A) $\frac{2MR^2}{5}$ (B) $\frac{7MR^2}{5}$
(C) $\frac{2MR^2}{3}$ (D) $\frac{7MR^2}{3}$

12. Moment of Inertia of the rectangle about AB is _____ cm^4 1 2 3 4



(A) 4.5 (B) 9
(C) 13.5 (D) 18

13. Motion of a particle along a straight line is given by $x = t^3 - 4.5t^2 + 5$, acceleration at time t = 5 seconds is 1 2 4 1
(A) $17.5 \frac{m}{s^2}$ (B) $30.0 \frac{m}{s^2}$
(C) $21.0 \frac{m}{s^2}$ (D) $23.5 \frac{m}{s^2}$

14. Work done by a horse while pulling a cart for a distance of 1km exerting a force of 1 kN will be 1 2 4 1
(A) 1000 J (B) 100 J
(C) 1000 kJ (D) 100 kJ

15. The centre of wheel rolling on a horizontal surface has a velocity of V_0. A point on the rim that is level with the centre will move at a rate of 1 2 4 1
(A) V_0 (B) $2V_0$
(C) $\sqrt{2}\,V_0$ (D) $\frac{V_0}{\sqrt{2}}$

16. In an elastic collision, the kinetic energy of the system _____ 1 1 4 1
(A) Increases (B) Decreases
(C) Remains Constant (D) First increases then reduces

17. The circumferential velocity of 12m/s. A wheel of radius 50cm. The angular velocity of the wheel is _____ rad/S 1 2 5 2
(A) 24 (B) 6
(C) 60 (D) 15

18. Change in Kinetic energy of a rigid body due to linear motion and rotary motion are 400J and 800J respectively. The work done by the rigid body is 1 2 5 2
(A) 1200 J (B) 400 J
(C) 600 J (D) 800 J

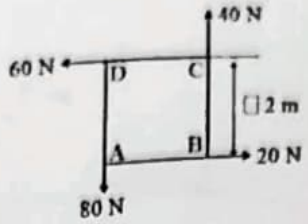
19. A particle performs uniform circular motion with an angular momentum "L". If the frequency of the particles motion is doubled in and its kinetic energy halved, the angular momentum becomes 1 2 5 1
(A) 2L (B) L
(C) $\frac{L}{2}$ (D) $\frac{L}{4}$

20. A mechanism with five links has _____ instantaneous centres 1 1 5 1
(A) 5 (B) 10
(C) 15 (D) 20

PART – B (5 × 8 = 40 Marks)

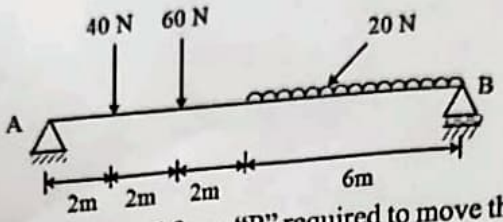
Answer **ALL** Questions

		Marks	BL	CO	PO
21. a.	Four forces of magnitudes 20N, 40N, 60N and 80N are acting respectively along the four sides of a square ABCD as shown. Locate the resultant force and its direction of action.	8	3	1	1

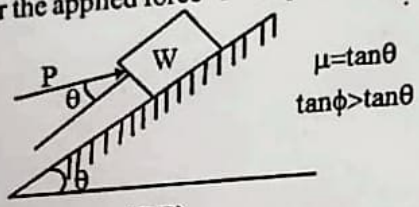


(OR)

b.	A simply supported beam is loaded as shown. Determine the support reactions at both ends.	8	3	2	2

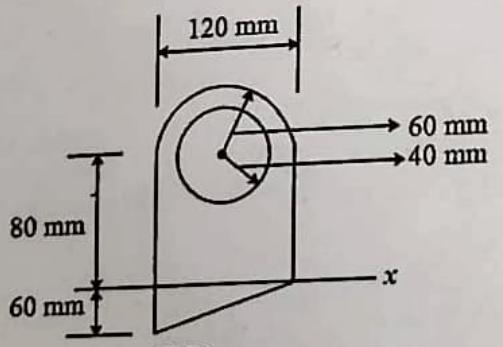


22. a.	Derive an expression for the applied force "P" required to move the block up the plane.	8	3	2	2



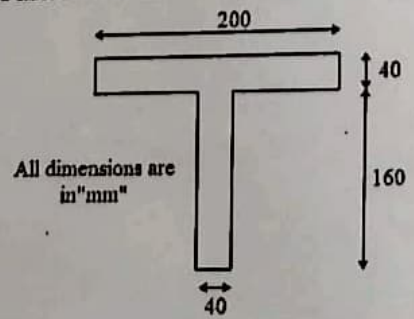
(OR)

b.	A ladder 4m long leans against a smooth vertical wall at an angle of 60° with the horizontal. The weight of the ladder is 700N. When a person of 600N stands on a rung 1.2m from the bottom of the ladder, the ladder is just about to slide. Determine the coefficient of friction between the ladder and the floor.	8	3	2	1
23. a.	A locate the centroid for the plane are shown.	8	2	3	2



(OR)

b.	Determine the moment of inertia about the centroid axes for the T section shown.	8	3	3	2

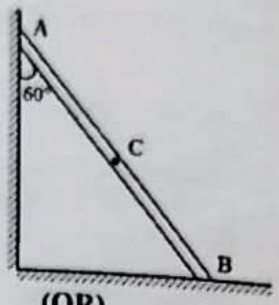


24. a. Two vehicles travel between two stations 50km apart. Both start at the same time from the same station. The first vehicle travels 50kmph whereas the second vehicle travels at 30kmph. If the first vehicle halts at the second station for 5 minutes and then starts with the same speed. Determine where the two vehicles will meet and at what time?

(OR)

b. Three perfectly elastic balls A,B and C of masses 2kg, 6kg and 12 kg are moving in the same direction with vehicles 12m/s, 4 m/s and 2 m/s respectively. If the ball A strikes ball B, which in turns strikes the ball C, Determine the velocity of the balls A, B & C after the collision.

25. a. Bar AB of length 5m slides in xy plane. The velocity of point A is 10m/s downwards and makes an angle 60° with the vertical. Determine the velocity of point B and mid-Point C.



(OR)

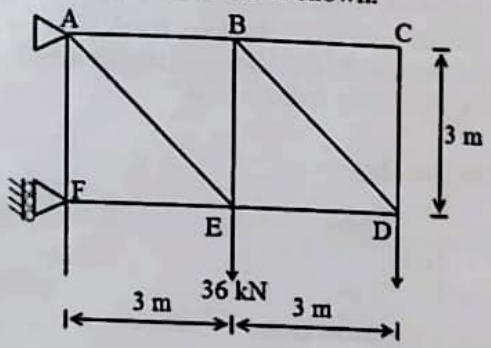
b. A sphere, a cylinder and a hoop each having the same mass and radius are released from rest on an incline. Determine the velocity of each body after it has rolled through a distance corresponding to a change in elevation 'h'.

PART – C (1 × 15 = 15 Marks)

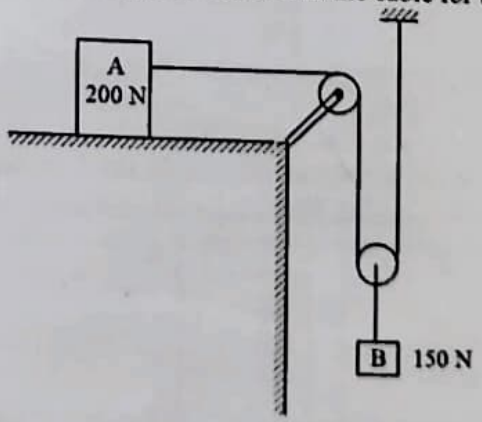
Answer **ANY ONE** Questions

Marks	BL	CO	PO
15	3	2	2

26. Find the forces in all members of the truss as shown.



27. Find the acceleration of the blocks and tension in the cable for the system shown. 15 4 5 2



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